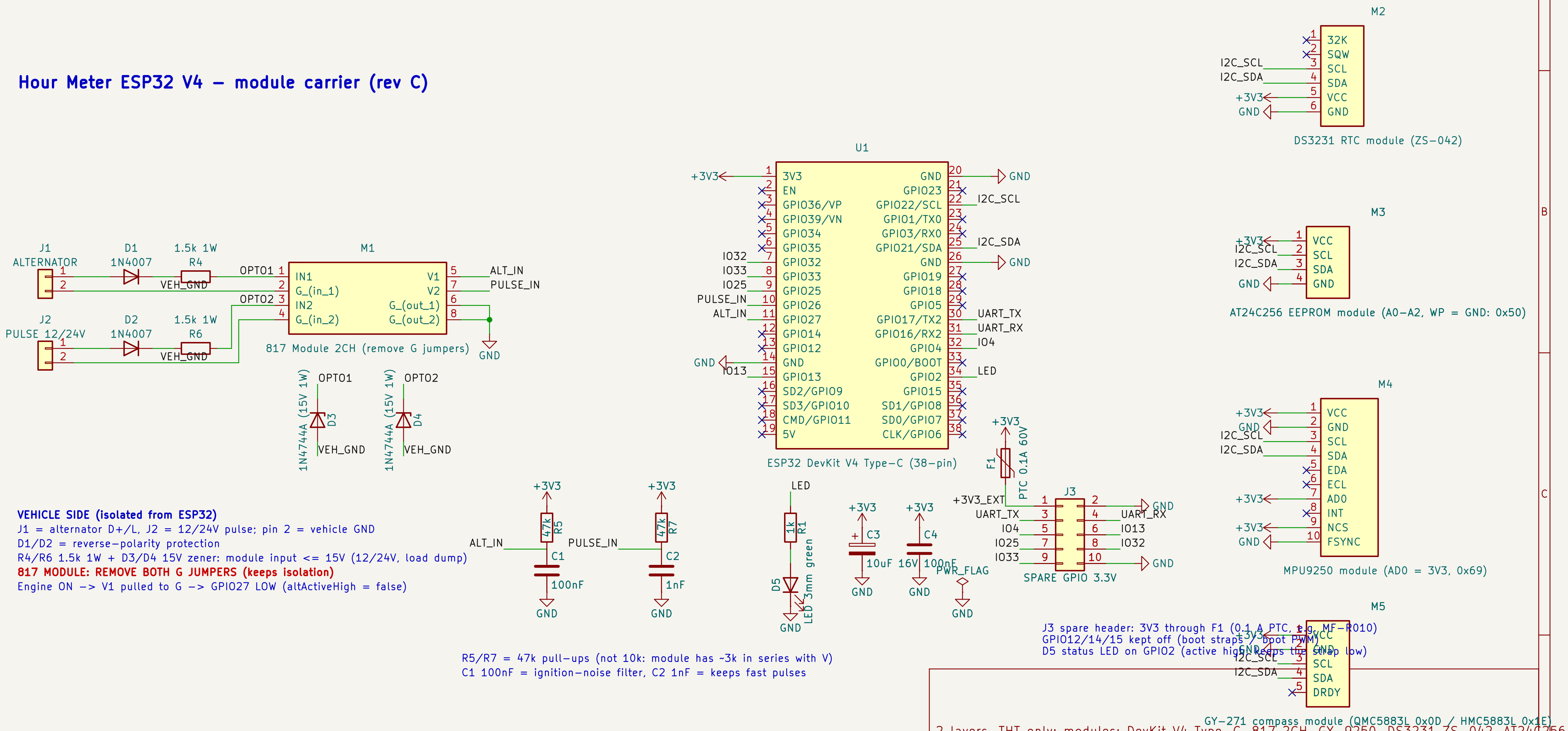
POWER: USB-C on the DevKit only (5V car charger in the vehicle). All modules run from the DevKit's 3V3

I2C bus (100 kHz, module pull-ups)
DS3231 0x68 | AT24C256 0x50 (A0-A2 = GND, WP = GND)
MPU9250 AD0 = 3V3 -> 0x69 (0x68 would clash with RTC)
ZS-042 RTC board: check its battery-charge resistor (CR2032)
Modules on 3V3: their I2C pull-ups stay at 3.3V (GPIOs are not 5V tolerant)
5V pin unused: nothing back-feeds the USB port

Hour Meter ESP32 V4 – module carrier (rev C)



VEHICLE SIDE (isolated from ESP32)
J1 = alternator D+/L, J2 = 12/24V pulse; pin 2 = vehicle GND
D1/D2 = reverse-polarity protection
R4/R6 1.5k 1W + D3/D4 15V zener: module input <= 15V (12/24V, load dump)
817 MODULE: REMOVE BOTH G JUMPERS (keeps isolation)
Engine ON -> V1 pulled to G -> GPIO27 LOW (altActiveHigh = false)

R5/R7 = 47k pull-ups (not 10k: module has ~3k in series with V)
C1 100nF = ignition-noise filter, C2 1nF = keeps fast pulses

J3 spare header: 3V3 through F1 (0.1 A PTC, e.g. MF-R010)
GPIO12/14/15 kept off (boot straps / boot PWM)
D5 status LED on GPIO2 (active high: keeps the strap low)

Compass: the GY-271 sits in the bottom strip of the board,
far from the DevKit (USB current, Wi-Fi) and the RTC's CR2032.
DRDY is not used: the firmware polls it.

2 layers, THT only; modules: DevKit V4 Type-C, 817 2CH, GY-9250, DS3231 ZS-042, AT24C256

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File: hour-meter-esp32-v4-compass.kicad_sch

Title: Hour Meter ESP32 V4 module carrier

Size: A4
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Rev: C
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